

The empirical recurrence for OEIS A254486: recovering a conjecture that is not written down

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Abstract

OEIS A254486 records “Empirical recurrence of order 66”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 219$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 66, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 66 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A254486 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum plus antidiagonal minimum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

19456, 278848, 3912576, 50028672, 640020096, 8348864128, 109168144000, 1413241384576, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:43:06, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1632$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 219$ of the 1632, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 219$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 438$ exact terms of the model returns order 66 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	18	c_{18}	615571959840768	c_{35}	283464896709609281028096	c_{52}	38761
c_2	−33	c_{19}	−2085554246123520	c_{36}	2883296282817395859390464	c_{53}	−93573
c_3	−496	c_{20}	12020148877656064	c_{37}	−8251610097073119228329984	c_{54}	670712
c_4	2048	c_{21}	−25650223296544768	c_{38}	9517522376702267559510016	c_{55}	−10406
c_5	−23040	c_{22}	−2925147606482944	c_{39}	−26525127858830680162041856	c_{56}	−27250
c_6	369664	c_{23}	323218347976032256	c_{40}	−76417131511611788042960896	c_{57}	797774
c_7	−4362240	c_{24}	−2862141467042775040	c_{41}	224220621549483387349106688	c_{58}	−40124
c_8	11141120	c_{25}	9064818336960872448	c_{42}	−138566250661159412342718464	c_{59}	560167
c_9	73252864	c_{26}	−47123129377723777024	c_{43}	156407821625414811553628160	c_{60}	401092
c_{10}	−414810112	c_{27}	115602044514438479872	c_{44}	528822994963760403848364032	c_{61}	−104977
c_{11}	6203310080	c_{28}	−103267170019698540544	c_{45}	−1579244354468301013073788928	c_{62}	519934
c_{12}	−39720058880	c_{29}	510687438964281835520	c_{46}	−52835902245681846120611840	c_{63}	396140
c_{13}	167520501760	c_{30}	2181359768377545981952	c_{47}	2241931681826156891089141760	c_{64}	−79228
c_{14}	−821476458496	c_{31}	−5415276951475475972096	c_{48}	−2133470729630848314643578880	c_{65}	158456
c_{15}	776122400768	c_{32}	−16560939979791681978368	c_{49}	3762328266368177820332457984	c_{66}	−79228
c_{16}	5126513229824	c_{33}	30368208010623202623488	c_{50}	1674230033904741056902070272		
c_{17}	−97350802472960	c_{34}	−96692871156445967548416	c_{51}	−3691832779511899756582404096		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 66, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $66 + 4$ terms removed returns order 66 as well, so $\deg q_{\min} = 66 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $66 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 66 that this sequence satisfies — and that family turns out to have exactly one member.

References

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